



Patient Understanding of Oncologic Radiology Reports: Is Access to Electronic Medical Records Helpful?

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Abstract

Access to electronic medical record (EMR) patient portals made it easier for patients to quickly acquire the results of their radiology studies. However, there is little research on how well oncology patients understand the findings of radiology reports presented in the online portal without patient-physician discussion. This study assessed oncology patients' confidence and accuracy in interpreting radiology reports either with or without layman translations. A survey based on a radiology report was administered to oncology patients and caregivers. Two versions of the radiological report were randomly distributed, either a standard report or one with layman translations to evaluate participant understanding and accuracy of interpreting radiological results. Among 85 participants, a majority (67.8%) reported wanting patient portal access to radiological reports, yet less than a quarter (21.2%) felt confident in reading and interpreting radiological reports. Univariate binary logistic regression models showed that participants who read the lay report were 8 times more likely to find the radiology report easy to read. This research demonstrated that the inclusion of layman translation of standard radiology reports improves oncology patients' and caregivers' understanding of such reports with statistically significant and clinically meaningful increases in readability.

Keywords Radiological reports · Oncology · Radiology

Introduction

Access to electronic medical record (EMR) patient portals has made it easier than ever for patients to acquire reports of their radiology studies. However, there is little research on how well oncology patients understand the findings of radiology reports presented in the online portal without patient-physician discussion. Thus, there are many unanswered questions about the value, benefit, or potentially negative implications of patient access to radiology reports.

Currently, radiology reports are written to be interpreted by the ordering physician and consulting providers. Therefore, such reports contain complex anatomic and pathologic language using polysyllabic, specialized medical terms to facilitate communication among healthcare providers. One exception to this style of reporting test results is due to a 1999 law that mandated timely release of mammography results directly to patients with results written in terms easily understood by a lay person [1]. Currently, there are increasing efforts to improve open access to health information, including radiological reports, under the 21st Century Cures Act [2]. However, there is evidence that patients' understanding, especially their diagnostic confidence, differs from the radiologists' interpretations of reports written in the standard format; this potential misinterpretation may contribute to patient confusion and anxiety over results [3]. Additionally, it has been shown that as the complexity of imaging modality increases from ultrasound and plain film radiographs to computed tomography (CT) and magnetic resonance imaging (MRI), patient understanding of results decreases [4].

Benefits of patient access to radiology reports include increased awareness of one's own health and improved shared decision-making between patient and provider. Conversely,

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potential consequences include increased patient anxiety and confusion, and more time-consuming physician–patient interactions [3]. A study by Fang et al. found that radiologists in general are open to additional interactions with patients to explain imaging results directly to patients, and Lee et al. found that with increased patient access to EMR records, radiologists have greater opportunities for communication with patients [5]. However, physicians do not always have time for additional patient appointments, and there is value in providing patients with information not only in a timely manner but in easily understandable language.

The main aim of this study was to assess oncology patients' and caregivers' confidence and accuracy in interpreting radiology reports, where the reports were presented either with or without layman translations. Secondly, this study assessed patient and caregiver confidence in previous experience reading and understanding radiological reports and their opinions on access to radiological results through an online portal.

Research Methods

Recruitment

This prospective, randomized study was conducted with the approval of the Queen's Medical Center (Honolulu, HI, USA) institutional review board (RA-2021–0024). Surveys were collected at a community cancer center located in Honolulu, Hawaii between April 2021 and June 2021. Participants, either oncology patients or their caregivers, were approached in the waiting room prior to their appointment and offered the opportunity to participate in the study. One survey was distributed per household to either the patient or an accompanying caregiver.

Survey Design

All consenting participants were randomized to read either a standard radiology report (standard report) or the standard radiology translated into layman terms (layman report). The radiology report is based on a computed tomography scan of the chest with both versions having findings and impressions sections. The standard radiology report was an actual de-identified radiological report. The supplemented layman terms had been previously reviewed by a radiologist and a medical oncologist to ensure information consistency and accuracy. The reading comprehension level of the layman report was checked using multiple readability tests to ensure readability at less than a 5th grade level. The survey items were composed of 10 demographic questions, 10 baseline opinion questions on access to radiology reports, and 7 questions assessing interpretation of the radiological results including two open ended

questions (Supplementary information 1). A Likert-type scale was used from “strongly agree” to “strongly disagree” for the opinion questions and questions assessing interpretation of the report. Randomization occurred by alternating the two versions of the radiology report. No personal identifiable information was collected. The survey was administered via Google Forms or hardcopy.

Data Analysis

The primary outcome measured was the participants' confidence in interpreting the radiology report. While there is no published literature comparing standard radiologic reports to reports with layman interpretations, we assumed a 30% difference in confidence between the two groups. With a level of significance (alpha) equal to 5%, we needed 72 patients to have 80% power for testing a 10% margin of superiority. To account for a 10% drop-out rate, the sample size goal was 80 participants.

Nonparametric descriptive statistics were used to evaluate participant characteristics. A 2-sided $p < 0.05$ was considered statistically significant. Binary logistic regression models for agreement with statements on (1) report readability, (2) recognition of medical terms, and (3) identification of new growths was built to obtain odds ratios (OR) and 95% confidence intervals (CI) adjusting for age, income, education level, patient vs. caregiver, and confidence in navigating the internet. Statistical analyses were performed with SPSS version 27.0 (IBM Corp).

Results

Study Population

In total, 85 participants completed the survey, of whom 69 were patients and 16 were caregivers or preferred to not answer. Table 1 summarizes the demographic information of the participants. The majority of participants were older than 65 years (63.5%) and retired (57.6%). A plurality were diagnosed with cancer within 2 years of completing the survey (41.2%). Over 80% of participants had some college education, and about half (48.2%) had an annual household income of at least \$60,000. The group with the layman report was slightly older; otherwise, there were no significant differences between the groups. Fifty-nine percent of participants had no experience reading radiological reports, and only 14.1% of participants had healthcare work experience. Nearly 70% of the respondents felt confident in their ability to navigate the internet.

Patient Preferences on Radiology Reports

The majority of participants (67.8%) strongly agreed or agreed that they want patient portal access to radiological

Table 1 Demographic information of the participants. Demographic table

	Standard report <i>n</i> (%)	Layman report <i>n</i> (%)
Gender		
M	16 (37)	19 (46)
F	27 (63)	22 (54)
Patient vs. caregiver		
Patient	38 (86)	31 (76)
Caregiver	6 (14)	10 (24)
Age*		
30 or younger	0 (0)	3 (7)
30–44	2 (5)	0 (0)
45–64	20 (45)	6 (15)
65–79	18 (41)	25 (61)
80 or older	4 (9)	7 (17)
Educational level		
Did not graduate from high school	5 (12)	6 (17)
Some college/college graduate	29 (69)	26 (72)
Post-graduate training	8 (19)	4 (11)
Time from cancer diagnosis		
Less than 1 year	8 (22)	10 (34)
1–2 years ago	14 (38)	3 (10)
3–5 years ago	6 (16)	9 (31)
More than 5 years ago	9 (24)	7 (24)
Annual income		
< \$30,000	7 (19)	8 (27)
\$30,000–59,000	6 (16)	5 (17)
\$60,000–89,000	11 (30)	5 (17)
> \$90,000	13 (35)	12 (39)
Employment status		
Retired	27 (64)	22 (54)
Disabled	3 (7)	3 (7)
Working full-time/part-time	12 (29)	16 (39)

**p*-value < 0.05

reports yet less than a quarter (21.2%) of respondents felt confident in reading and interpreting radiological reports (Fig. 1). Accessing results quickly was important to most participants (82.3%), but 72.9% would still prefer waiting for a doctor's explanation even if it meant a delay in results. Fifty-three percent of participants reported feeling anxious while waiting for results of their radiology scans and 63.5% of respondents reported that electronic access to the results of the radiology scans would have relieved that anxiety.

Standard Report vs. Layman Report

The respondents who read the layman report also found the modified medical terminology to be more familiar and were more often able to identify tumor progression (defined herein

as the presence of “new growths”). Participant characteristics such as age, income, education level, patient vs. caregiver, and confidence in navigating the internet were not significantly associated with ability to understand the radiology report or medical terms or to identify new growths (Table 2).

Univariate binary logistic regression models showed that participants who read the lay report were 8 times more likely to find the radiology report easy to read (Table 3). Similar significant results were seen with the layman report group being less likely to be unfamiliar with the medical terms (OR 0.29, 95% CI 0.11–0.77) and more likely to identify new growths (OR 4.69, 95% CI 1.38–15.98). These significant associations with lay language radiology reports persisted after adjustment for age, income, education level, patient vs. caregiver, and confidence in navigating the internet.

In the free response answers from participants who read the standard report, the use of medical terminology emerged as a point of confusion. When participants who read the standard radiological report were asked what the hardest part of understanding the report was, one participant stated “not understanding medical terms” while another said “all of it was hard to read, easy to get confused.” In contrast, answers to this question on the layman report were much more favorable with one participant stating that “it was pretty clear, no real trouble understanding it” and another found it “overall easy to understand.” These comments were typical of the responses received.

Discussion and Conclusion

This study demonstrated that the use of layman's terms in radiology report improves oncology patients' and caregivers' understanding of such reports. The difference in readability of the reports was statistically significant and clinically meaningful; 82% of participants reported easy readability with the layman report compared to only 33% with the standard reports. Additionally, participants were able to better identify the presence of “new growths” on the layman report than the standard report. This study demonstrates that patients and caregivers do not usually have the ability to understand the complex medical jargon written in standard radiology reports. This finding is consistent with other studies showing that patients have difficulty understanding standard reports and would prefer lay translations [6, 7]. With patient results now accessible through online portals, there is an even greater need to provide patients quickly with clinical interpretations that are simple and understandable.

We found that the vast majority of oncology patients and caregivers want patient portal access to radiological reports; however, fewer than a quarter of the respondents felt confident reading and interpreting radiological reports. Over half of the respondents felt anxious waiting for the

Access to Radiological Report Statements

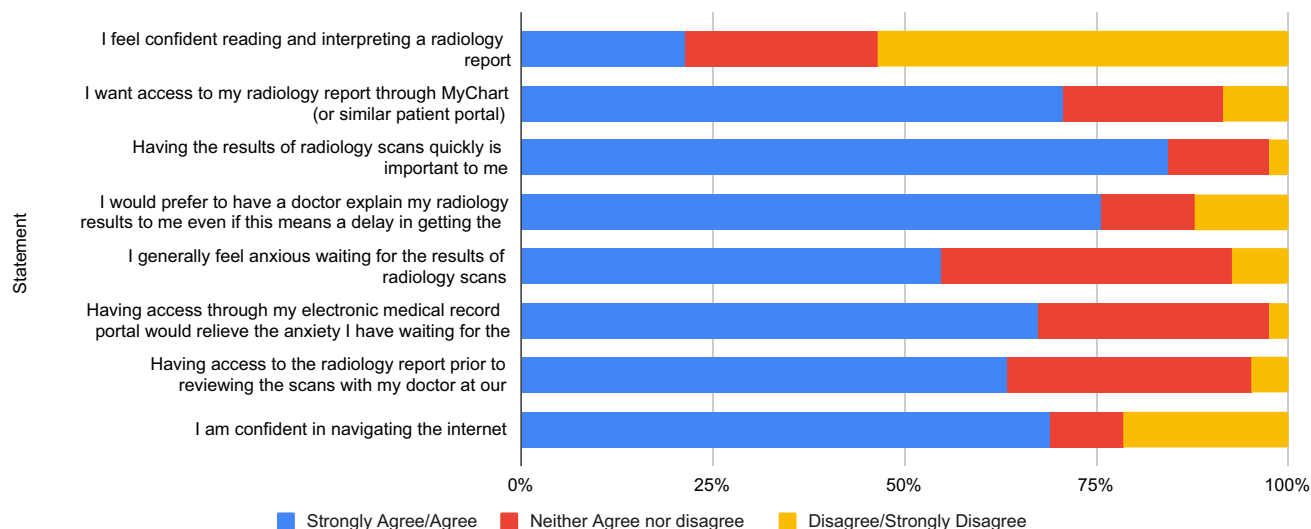


Fig. 1 Statements on access to radiology reports from patients or caregivers

Table 2 Understanding and comfortability of radiology report in Standard or Layman's terminology for cancer patients or caregivers

Statement		Agree <i>n</i> (%)	Neutral <i>n</i> (%)	Disagree <i>n</i> (%)	<i>p</i> value
I feel the language in the report was easy to read and understand	Standard	14 (33)	14 (33)	14 (33)	<0.0005
	Lay	31 (82)	6 (16)	1 (3)	
I am not familiar with the medical terms used in the report	Standard	22 (52)	15 (36)	5 (12)	0.009
	Lay	9 (24)	14 (38)	14 (38)	
The radiology report suggests there are new growths	Standard	24 (62)	11 (28)	4 (10)	0.027
	Lay	30 (88)	2 (6)	2 (6)	

Table 3 Univariate and multivariate binary logistic regression

	Univariate		Multivariate	
	OR (95% CI)	<i>p</i> value	OR (95% CI)	<i>p</i> value
I feel the language in the report was easy to read and understand	8.86 (3.13–25.09)	<0.0005	19.32 (4.16–89.76)	<0.0005
I am not familiar with the medical terms used in the report	0.29 (0.11–0.77)	0.012	0.23 (0.072–0.70)	0.010
The radiology report suggests there are new growths	4.69 (1.38–15.98)	0.014	5.05 (1.11–23.02)	0.037

results of the radiological scans and more than 60% felt having online access would relieve some of that anxiety. Separate research has shown that patients with increased anxiety are more likely to incorrectly interpret imaging results [8]. This creates an interesting paradigm where oncology patients want access to radiological results to relieve some “scanxiety” prior to provider appointments but yet do not feel confident interpreting the standard radiological reports. Lack of confidence on the part of patients may contribute to undue anxiety and also increase the chance of incorrect interpretation of the results [9].

As of April of 2021, under the 21st Century Cures Act, health-care providers and hospitals are required to release finalized imaging reports immediately to patients [2]. We found that the vast majority of participants (82.3%) want access to radiological reports quickly. However, 72.9% of oncology patients and caregivers also stated they would prefer to have a doctor explain radiology results even if this means a delay in getting the results. This divergence could be addressed by providing an interpretation of the results in an easier to understand format, and this study demonstrates that patients can understand radiological reports with layman interpretations more easily than standard reports.

This study confirms the benefit of including lay interpretations in radiology reports for cancer patients and their caregivers. Future directions include ongoing efforts to work directly with radiologists to form a standardized methodology to translate the complex radiological report to the level that is understandable by the average patient. Possible strategies to address ongoing patient concerns include having radiologists write an additional patient-focused interpretation or integrating technology such as the PORTER (Patient-Oriented Radiology Reporter) which acts to augment radiology reports with lay-language definitions linked to Wikipedia into the production of radiology reports [10, 11]. These approaches, while beneficial to patients, would undoubtedly add to the time and cost of generating reports. However, similar practice has successfully been implemented for mammograms results for several years. Release of mammogram results directly to the patient in words that are easy to understand allows for greater patient ownership of their breast health and involvement in their care [12], setting a precedent for expanding this practice to other studies.

Our study has several strengths. Rather than focus on a specific patient population, our sample included a variety of cancer types. We also considered responses from both patients and caregivers, groups that may have varying past experience with radiology reports. The diversity of this study population makes the findings more generalizable. There are also some limitations to this study. While the study met its accrual goal of 80 patients, enrollment was limited to one health care system which could limit the ability to extrapolate the findings to other health care systems. However, given the diversity of study participants and the magnitude of benefit seen with the lay report, it is not likely that a study population spanning multiple health-care systems would yield different findings. Another limitation could be the presence of a social desirability effect in which participants may have answered questions such as “the language was easy to understand” differently because they knew their answers were being recorded. Additionally, the majority of participants had at least some college education which is higher than the general population.

In summary, this study demonstrates that the inclusion of a layman translation with standard radiology reports improves oncology patients’ and caregivers’ understanding of such reports with statistically significant and clinically meaningful increases in readability. This study provides evidence for switching from the current practice of releasing standard radiological reports to patients through online portals to a more patient-focused incorporation of patient-centered layman interpretations of such reports. By carefully modifying our procedures and continuing to evaluate patient needs, the medical community will be able to continue in providing high quality patient care and satisfaction.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s13187-022-02204-5>.

Declarations

Conflict of Interest The authors declare no competing interests.

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